Milestone 1: Data Collection and Preparation (Weeks 1–2)

Objective

The objective of Milestone 1 was to design and implement an automated web scraping and preprocessing pipeline to collect military and economic indicators for over 140 countries from GlobalFirepower.com and transform them into a structured, analytics-ready dataset.

Module 1: Web Scraping Setup and Execution

Goal

To dynamically scrape country-level military metrics across 50+ indicators using predefined metric URLs and convert them into a structured dataset.

Metric URL Mapping

A dictionary was created to map each metric URL to a standardized column name. This allowed dynamic looping across all indicators.

```
Dict = {
    'https://www.globalfirepower.com/total-population-by-country.php':
'total_population',
    'https://www.globalfirepower.com/active-military-manpower.php':
'active_personnel',
    'https://www.globalfirepower.com/aircraft-total.php':
'total_military_aircraft',
    ...
}
```

This structure ensured scalability and avoided hardcoding individual scraping logic for each metric.

Web Scraping Logic

For each URL, the HTML content was fetched and parsed:

```
response = requests.get(u)
```

```
soup = BeautifulSoup(response.text, 'html.parser')
```

All relevant country links were filtered using:

```
if "country_id" in href:
```

This ensured that only country-specific records were processed.

Data Extraction

For each country entry, the following elements were extracted:

```
country_div = record.find("div", class_="longFormName")
value_div = record.find("div", class_="valueContainer")
```

Extracted fields:

- Country name
- Metric value

Cleaning During Extraction

Values were cleaned using regex to remove non-numeric characters:

```
regex = "[^0-9]"
value = (re.sub(regex, "", value))
```

This removed:

- Commas
- Symbols
- Units
- Special characters

Values were then converted into integers before storage.

Structured Storage

Each metric record was appended into a structured list:

```
final_result.append({
    "country": country,
    "metric": name,
    "value": int(value)
```

```
})
```

DataFrame Creation

After collecting all records, a pandas DataFrame was created:

```
df = pd.DataFrame(final_result)
```

At this stage, the dataset was in long format (country, metric, value).

Pivot Transformation (Long → Wide)

To make the dataset dashboard-ready, it was reshaped:

```
df = df.pivot(
    index="country",
    columns="metric",
    values="value"
)
df = df.reset_index()
```

This produced:

- One row per country
- One column per metric

Missing Value Inspection

Missing values were analyzed:

```
df.isnull().sum()
```

Unnecessary columns were removed where needed:

```
df.drop('total_naval_fleet_tonnage_mt', axis=1, inplace=True)
```

Deliverables

- Scraping script handling 50+ metric URLs
- Structured DataFrame for 140+ countries
- Country-level unified dataset

Module 2: Data Cleaning and Standardization

Goal

To transform raw scraped data into a consistent, numeric, and dashboard-ready format.

Cleaning & Structuring Process

The following preprocessing steps were applied:

- Removed formatting symbols (commas, %, +, special characters)
- Standardized column naming conventions
- Ensured consistent numeric data types
- Identified and handled missing values
- Validated dataset integrity

All cleaning operations ensured compatibility with downstream KPI engineering and visualization tools such as Tableau and Power BI.

Output Files

- Cleaned structured dataset ready for export
- Pivoted country-level dataset for analysis